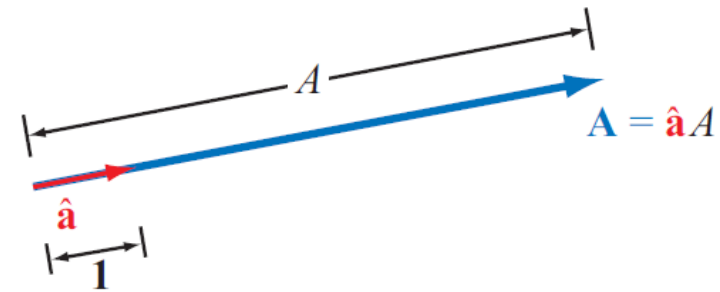


Vectors and Coordinates

Laws of Vector Algebra

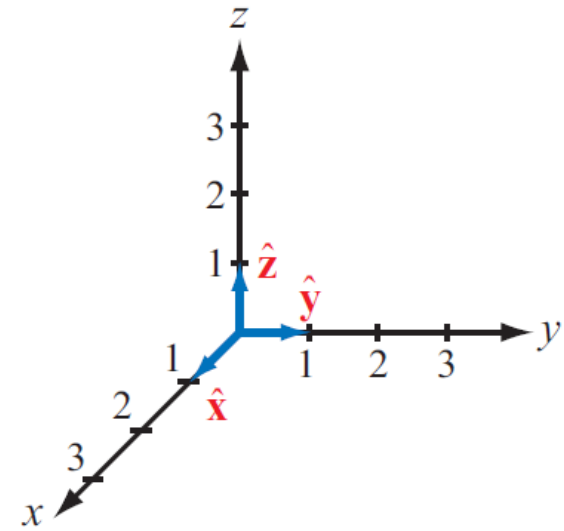


$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{a}}A$$

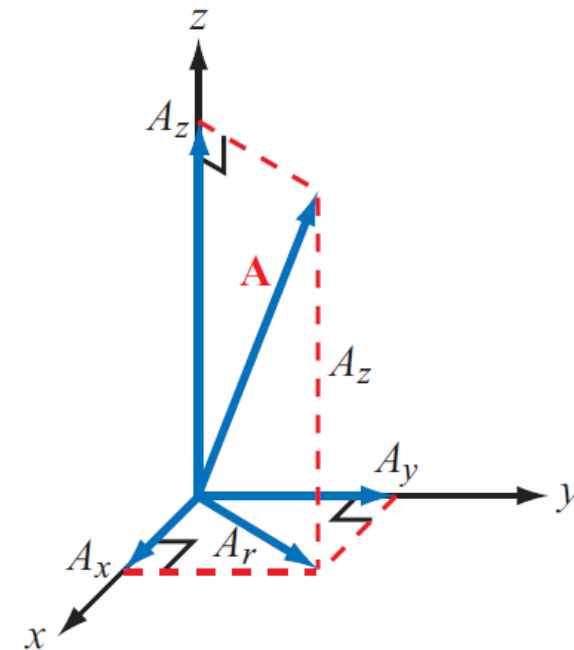
$$\mathbf{A} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$$

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = \frac{\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$



(a) Base vectors



(b) Components of $\mathbf{A}$

Properties of Vector Operations

Equality of Two Vectors

$$\mathbf{A} = \hat{\mathbf{a}}A = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z, \quad (3.6a)$$

$$\mathbf{B} = \hat{\mathbf{b}}B = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.6b)$$

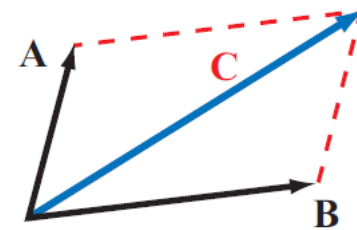
then $\mathbf{A} = \mathbf{B}$ if and only if $A = B$ and $\hat{\mathbf{a}} = \hat{\mathbf{b}}$, which requires that $A_x = B_x$, $A_y = B_y$, and $A_z = B_z$.

Equality of two vectors does not necessarily imply that they are identical; in Cartesian coordinates, two displaced parallel vectors of equal magnitude and pointing in the same direction are equal, but they are identical only if they lie on top of one another.

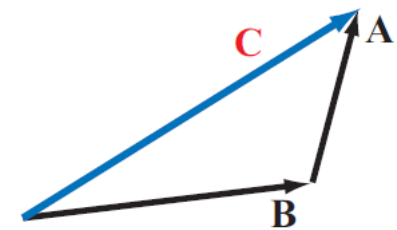
Properties of Vector Operations

Addition

$$\begin{aligned}\mathbf{C} = \mathbf{A} + \mathbf{B} &= (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) + (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z) \\ &= \hat{\mathbf{x}}(A_x + B_x) + \hat{\mathbf{y}}(A_y + B_y) + \hat{\mathbf{z}}(A_z + B_z) \\ &= \hat{\mathbf{x}}C_x + \hat{\mathbf{y}}C_y + \hat{\mathbf{z}}C_z,\end{aligned}$$



(a) Parallelogram rule



(b) Head-to-tail rule

Commutative property

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

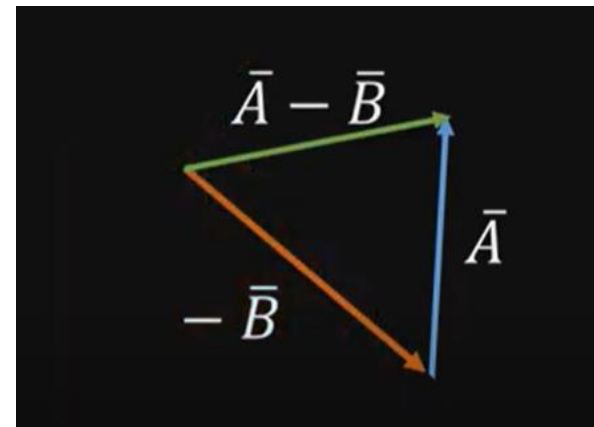
Figure 3-3: Vector addition by (a) the parallelogram rule and (b) the head-to-tail rule.

Properties of Vector Operations

Subtraction

$$\begin{aligned}\mathbf{D} = \mathbf{A} - \mathbf{B} &= \mathbf{A} + (-\mathbf{B}) \\ &= \hat{\mathbf{x}}(A_x - B_x) + \hat{\mathbf{y}}(A_y - B_y) + \hat{\mathbf{z}}(A_z - B_z).\end{aligned}$$

Graphically, the same rules used for vector addition are also applicable to vector subtraction; the only difference is that the arrowhead of $(-\mathbf{B})$ is drawn on the opposite end of the line segment representing the vector $\mathbf{B}$ (i.e., the tail and head are interchanged).



Position & Distance Vectors

Position Vector: From origin to point P

$$\mathbf{R}_1 = \overrightarrow{OP_1} = \hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1$$

$$\mathbf{R}_2 = \overrightarrow{OP_2} = \hat{\mathbf{x}}x_2 + \hat{\mathbf{y}}y_2 + \hat{\mathbf{z}}z_2$$

Distance Vector: Between two points

$$\begin{aligned}\mathbf{R}_{12} &= \overrightarrow{P_1P_2} \\ &= \mathbf{R}_2 - \mathbf{R}_1 \\ &= \hat{\mathbf{x}}(x_2 - x_1) + \hat{\mathbf{y}}(y_2 - y_1) + \hat{\mathbf{z}}(z_2 - z_1)\end{aligned}$$

the distance d between P_1 and P_2 equals the magnitude of $\mathbf{R}_{12}$:

$$\begin{aligned}d &= |\mathbf{R}_{12}| \\ &= [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}. \quad (3.12)\end{aligned}$$

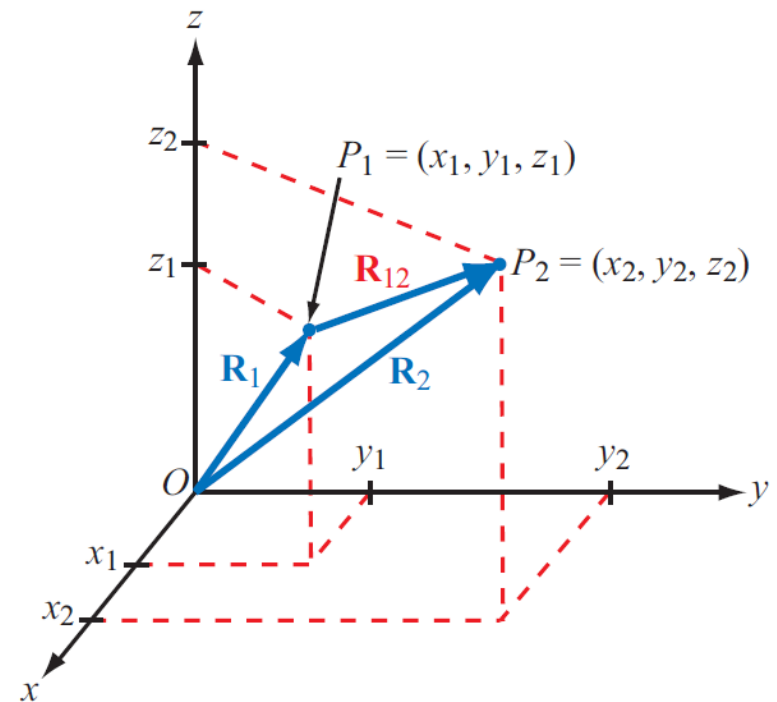


Figure 3-4: Distance vector $\mathbf{R}_{12} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$, where $\mathbf{R}_1$ and $\mathbf{R}_2$ are the position vectors of points P_1 and P_2 , respectively.

Vector Multiplication: Scalar Product or "Dot Product"

If $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$, then

$$\mathbf{A} \cdot \mathbf{B} = (\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z) \cdot (\hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z).$$

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

Hence:

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z.$$

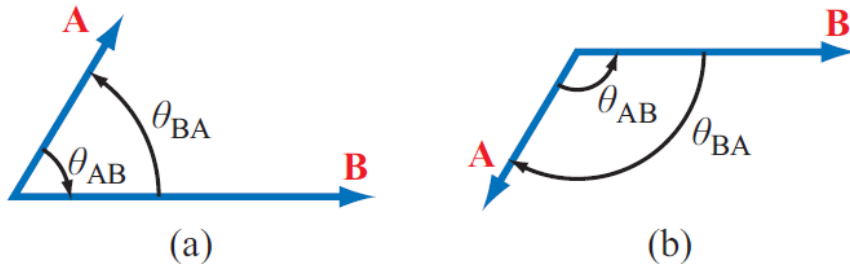


Figure 3-5: The angle θ_{AB} is the angle between $\mathbf{A}$ and $\mathbf{B}$, measured from $\mathbf{A}$ to $\mathbf{B}$ between vector tails. The dot product is positive if $0 \leq \theta_{AB} < 90^\circ$, as in (a), and it is negative if $90^\circ < \theta_{AB} \leq 180^\circ$, as in (b).

$$A = |\mathbf{A}| = \sqrt{\mathbf{A} \cdot \mathbf{A}}$$

$$\theta_{AB} = \cos^{-1} \left[\frac{\mathbf{A} \cdot \mathbf{B}}{\sqrt{\mathbf{A} \cdot \mathbf{A}} \sqrt{\mathbf{B} \cdot \mathbf{B}}} \right]$$

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} \quad (\text{commutative property}),$$

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C} \quad (\text{distributive property})$$

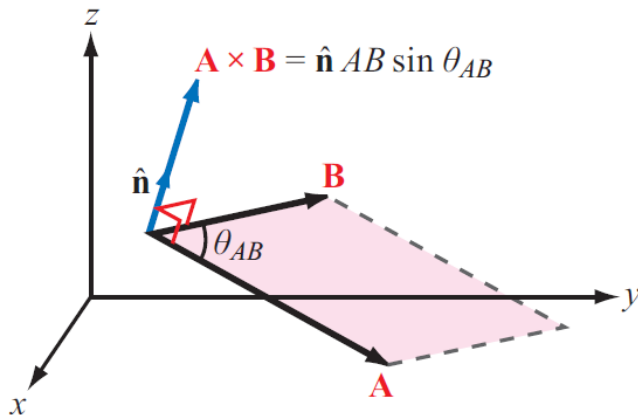
$$\begin{aligned} \hat{\mathbf{x}} \cdot \hat{\mathbf{x}} &= \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1, \\ \hat{\mathbf{x}} \cdot \hat{\mathbf{y}} &= \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0. \end{aligned}$$

Vector Multiplication: Vector Product or "Cross Product"

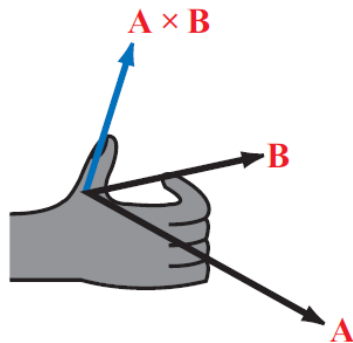
If $\mathbf{A} = (A_x, A_y, A_z)$ and $\mathbf{B} = (B_x, B_y, B_z)$,

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.$$



(a) Cross product



(b) Right-hand rule

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A} \quad (\text{anticommutative})$$

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C} \quad (\text{distributive})$$

$$\mathbf{A} \times \mathbf{A} = 0$$

$$\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}, \quad \hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}, \quad \hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}. \quad (3.25)$$

Note the cyclic order ($xyzxyz \dots$). Also,

$$\hat{\mathbf{x}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0. \quad (3.26)$$

Triple Products

Scalar Triple Product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}).$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

Vector Triple Product

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}),$$

which is known as the “bac-cab” rule.

Example 3-2: Vector Triple Product

Given $\mathbf{A} = \hat{x} - \hat{y} + \hat{z}2$, $\mathbf{B} = \hat{y} + \hat{z}$, and $\mathbf{C} = -\hat{x}2 + \hat{z}3$, find $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ and compare it with $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

Solution:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{x}3 - \hat{y} + \hat{z}$$

and

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -3 & -1 & 1 \\ -2 & 0 & 3 \end{vmatrix} = -\hat{x}3 + \hat{y}7 - \hat{z}2.$$

A similar procedure gives $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \hat{x}2 + \hat{y}4 + \hat{z}$.

Hence:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$$

Cartesian Coordinate System

Differential length vector

$$d\mathbf{l} = \hat{\mathbf{x}} dl_x + \hat{\mathbf{y}} dl_y + \hat{\mathbf{z}} dl_z = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz, \quad (3.34)$$

where $dl_x = dx$ is a differential length along $\hat{\mathbf{x}}$, and similar interpretations apply to $dl_y = dy$ and $dl_z = dz$.

Differential area vectors

$$ds_x = \hat{\mathbf{x}} dl_y dl_z = \hat{\mathbf{x}} dy dz \quad (\text{y-z plane}), \quad (3.35a)$$

with the subscript on ds denoting its direction. Similarly,

$$ds_y = \hat{\mathbf{y}} dx dz \quad (\text{x-z plane}), \quad (3.35b)$$

$$ds_z = \hat{\mathbf{z}} dx dy \quad (\text{x-y plane}). \quad (3.35c)$$

A **differential volume** equals the product of all three differential lengths:

$$dV = dx dy dz. \quad (3.36)$$

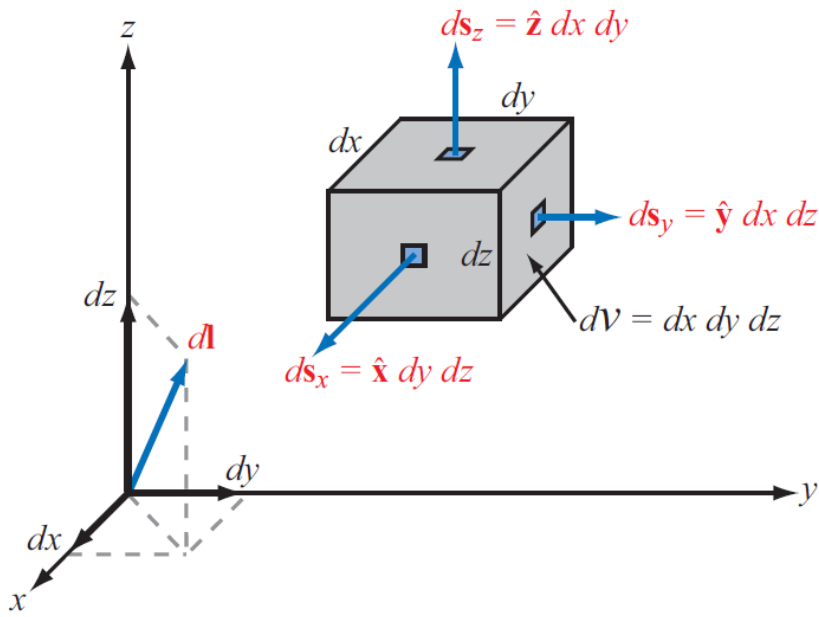
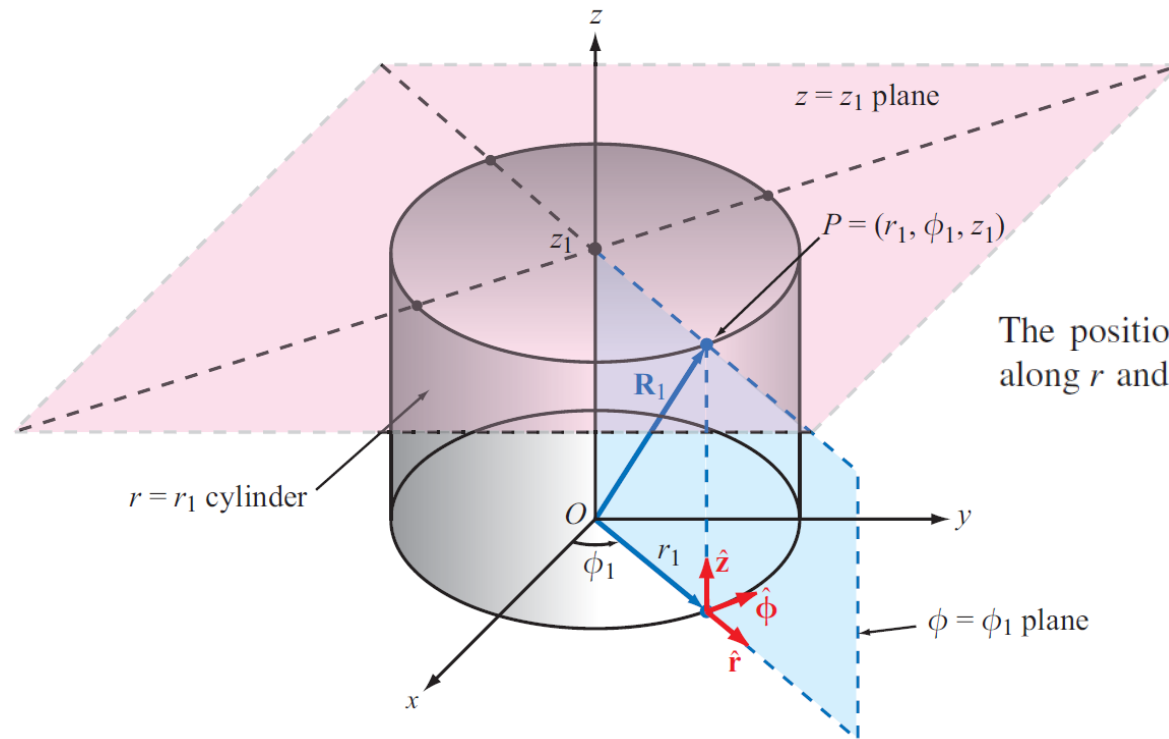


Table 3-1: Summary of vector relations.

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1,$ for $P = (x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1,$ for $P = (r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1,$ for $P = (R_1, \theta_1, \phi_1)$
Base vectors properties	$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}} \cdot \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_x B_x + A_y B_y + A_z B_z$	$A_r B_r + A_\phi B_\phi + A_z B_z$	$A_R B_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz$	$\hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz$	$\hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{\mathbf{x}} dy dz$ $ds_y = \hat{\mathbf{y}} dx dz$ $ds_z = \hat{\mathbf{z}} dx dy$	$ds_r = \hat{\mathbf{r}} r d\phi dz$ $ds_\phi = \hat{\boldsymbol{\phi}} dr dz$ $ds_z = \hat{\mathbf{z}} r dr d\phi$	$ds_R = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi$ $ds_\phi = \hat{\boldsymbol{\phi}} R dR d\theta$
Differential volume $dV =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

Cylindrical Coordinate System



The position vector $\vec{OP}$ shown in Fig. 3-9 has components along r and z only. Thus,

$$\mathbf{R}_1 = \vec{OP} = \hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1. \quad (3.40)$$

The mutually perpendicular base vectors are $\hat{\mathbf{r}}$, $\hat{\phi}$, and $\hat{\mathbf{z}}$, with $\hat{\mathbf{r}}$ pointing away from the origin along r , $\hat{\phi}$ pointing in a direction tangential to the cylindrical surface, and $\hat{\mathbf{z}}$ pointing along the vertical. Unlike the Cartesian system, in which the base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are independent of the location of P , in the cylindrical system both $\hat{\mathbf{r}}$ and $\hat{\phi}$ are functions of ϕ .

$$\begin{aligned} -\infty &\leq z \leq +\infty \\ 0 &\leq r \leq +\infty \\ 0 &\leq \phi \leq 2\pi \end{aligned}$$

Cylindrical Coordinate System

The base unit vectors obey the following right-hand cyclic relations:

$$\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}, \quad \hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}, \quad (3.37)$$

and like all unit vectors, $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$, and $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$.

In cylindrical coordinates, a vector is expressed as

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z, \quad (3.38)$$

$$dl_r = dr, \quad dl_\phi = r d\phi, \quad dl_z = dz. \quad (3.41)$$

Note that the differential length along $\hat{\boldsymbol{\phi}}$ is $r d\phi$, not just $d\phi$. The differential length $d\mathbf{l}$ in cylindrical coordinates is given by

$$d\mathbf{l} = \hat{\mathbf{r}} dl_r + \hat{\boldsymbol{\phi}} dl_\phi + \hat{\mathbf{z}} dl_z = \hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz. \quad (3.42)$$

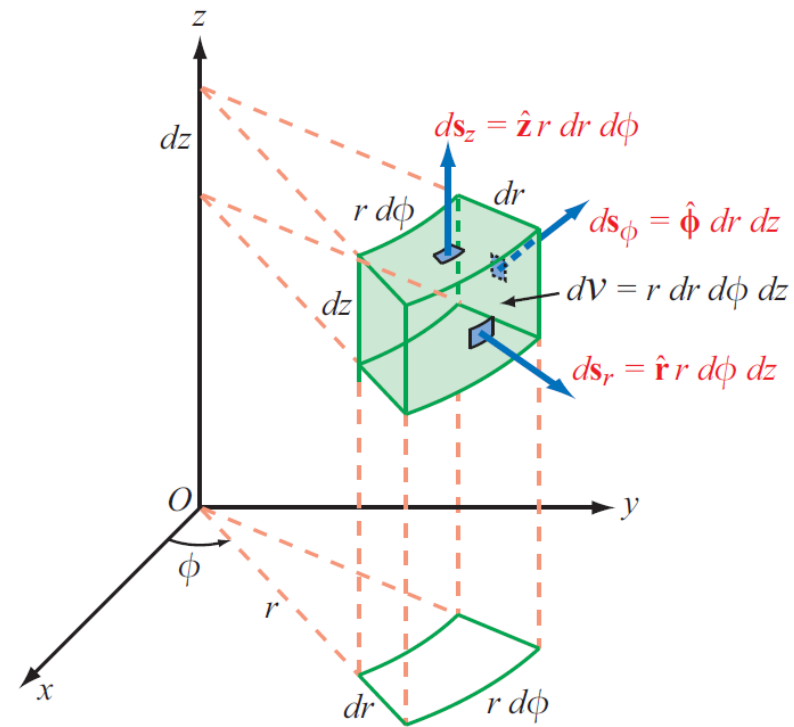


Figure 3-10: Differential areas and volume in cylindrical coordinates.

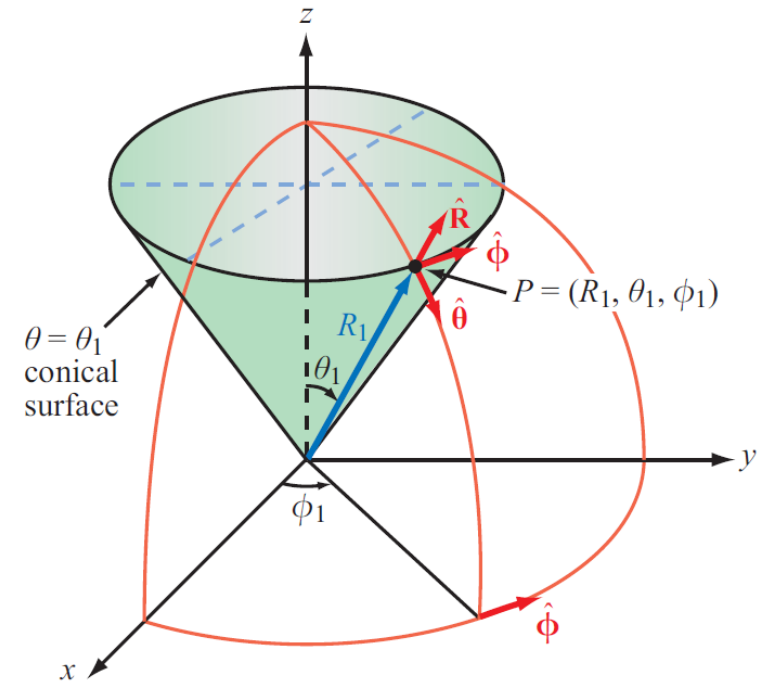
Spherical Coordinate

The range coordinate $\mathbf{R}$ which measures the distance from the origin to the point, describes a sphere of radius centered at the origin. The zenith angle θ is measured from the positive z-axis and it describes a conical surface with its apex at the origin, and the azimuth angle ϕ is the same as in cylindrical coordinates.

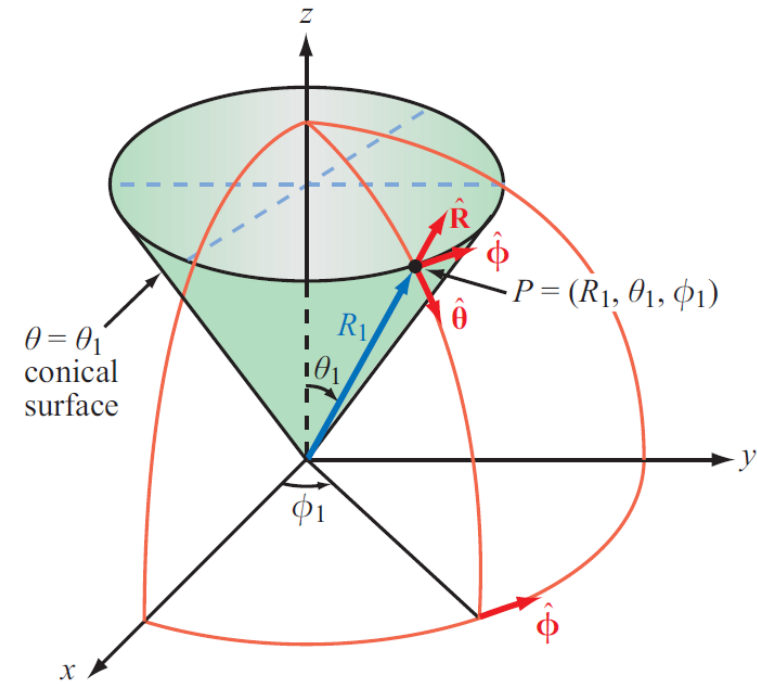
$$0 \leq R \leq +\infty$$

$$0 \leq \theta \leq \pi$$

$$0 \leq \phi \leq 2\pi$$



Spherical Coordinate



$$\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}, \quad \hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}. \quad (3.45)$$

A vector with components A_R , A_θ , and A_ϕ is written as

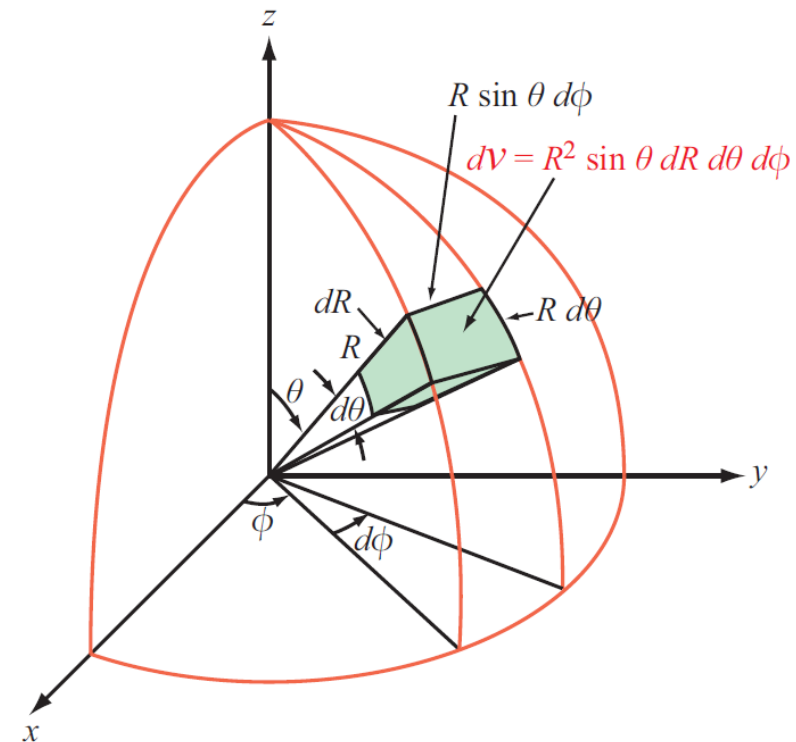
$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi, \quad (3.46)$$

and its magnitude is

$$|\mathbf{A}| = \sqrt[+]{\mathbf{A} \cdot \mathbf{A}} = \sqrt[+]{A_R^2 + A_\theta^2 + A_\phi^2}. \quad (3.47)$$

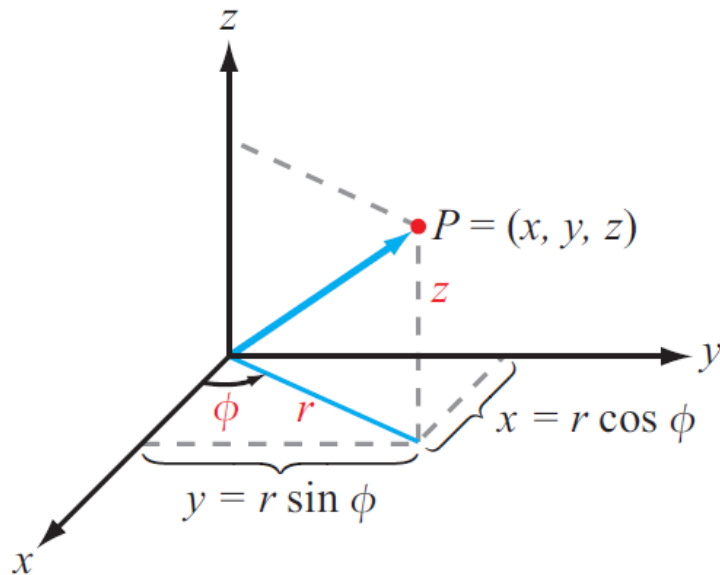
The position vector of point $P = (R_1, \theta_1, \phi_1)$ is simply

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{R}}R_1, \quad (3.48)$$



Coordinate Transformations: Coordinates

- To solve a problem, we select the coordinate system that best fits its geometry
- Sometimes we need to transform between coordinate systems



$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \left(\frac{y}{x} \right),$$

and the inverse relations are

$$x = r \cos \phi, \quad y = r \sin \phi.$$

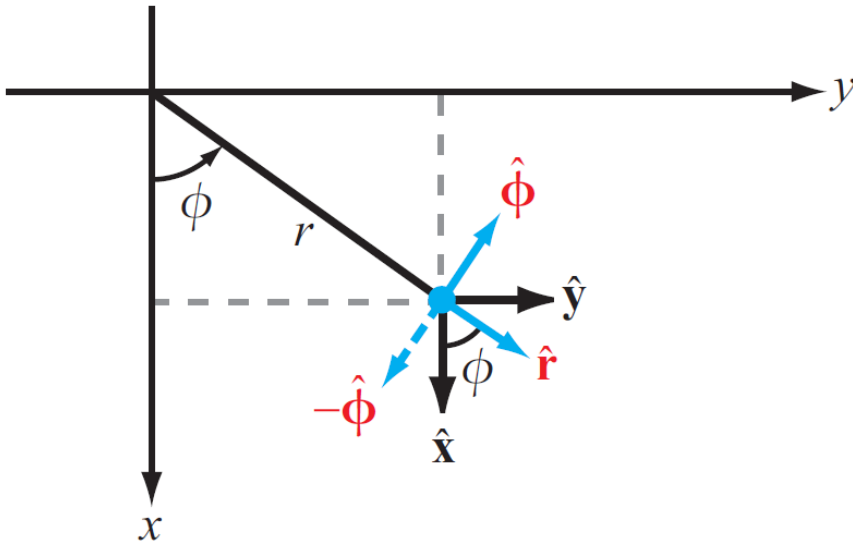
Figure 3-16: Interrelationships between Cartesian coordinates (x, y, z) and cylindrical coordinates (r, ϕ, z) .

Coordinate Transformations: Unit Vectors

$$\begin{aligned}\hat{\mathbf{r}} \cdot \hat{\mathbf{x}} &= \cos \phi, & \hat{\mathbf{r}} \cdot \hat{\mathbf{y}} &= \sin \phi, \\ \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{x}} &= -\sin \phi, & \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{y}} &= \cos \phi.\end{aligned}$$

$$\hat{\mathbf{r}} = \hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi.$$

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi.$$



$$\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi,$$

$$\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi.$$

Table 3-2: Coordinate transformation relations.

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$ $\hat{z} = \hat{z}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{z}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{R} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$ $\hat{\theta} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{x} = \hat{R} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{R} \sin \theta \sin \phi + \hat{\theta} \cos \theta \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{R} = \hat{r} \sin \theta + \hat{z} \cos \theta$ $\hat{\theta} = \hat{r} \cos \theta - \hat{z} \sin \theta$ $\hat{\phi} = \hat{\phi}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{r} = \hat{R} \sin \theta + \hat{\theta} \cos \theta$ $\hat{\phi} = \hat{\phi}$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

Distance Between 2 Points

$$\begin{aligned}d &= |\mathbf{R}_{12}| \\&= [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}. \quad (3.66)\end{aligned}$$

$$\begin{aligned}d &= [(r_2 \cos \phi_2 - r_1 \cos \phi_1)^2 \\&\quad + (r_2 \sin \phi_2 - r_1 \sin \phi_1)^2 + (z_2 - z_1)^2]^{1/2} \\&= [r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2} \\&\quad \text{(cylindrical)}. \quad (3.67)\end{aligned}$$

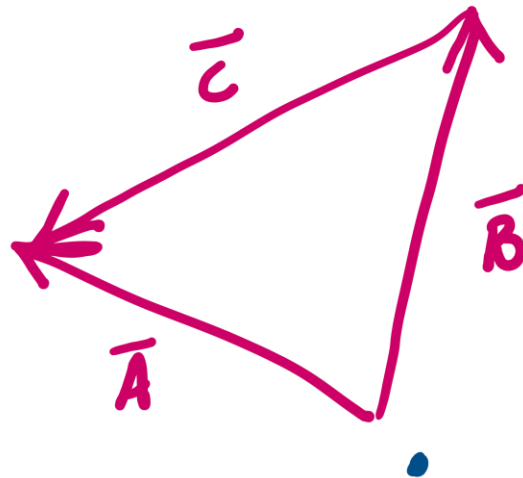
$$\begin{aligned}d &= \{R_2^2 + R_1^2 - 2R_1 R_2 [\cos \theta_2 \cos \theta_1 \\&\quad + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2} \\&\quad \text{(spherical)}. \quad (3.68)\end{aligned}$$

Tips

- If you need to add to subtract vector
 - Use cartesian coordinates
 - Much more complicated in the other systems
- For Dot and Cross Multiplication
 - Doesn't matter all three systems are very similar

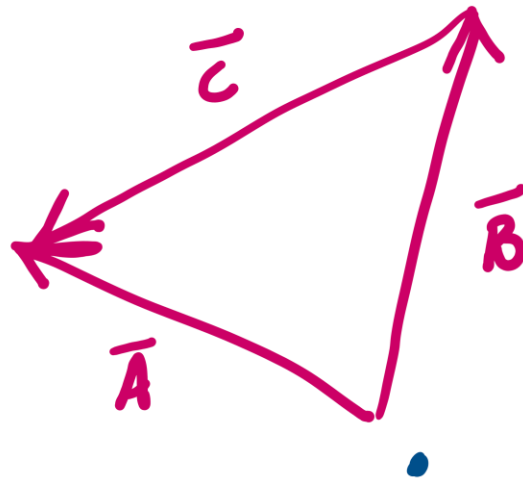
PROBLEM ①

WRITE THE EXPRESSION FOR THE
RELATIONSHIP BETWEEN THE
THREE VECTORS



PROBLEM ①

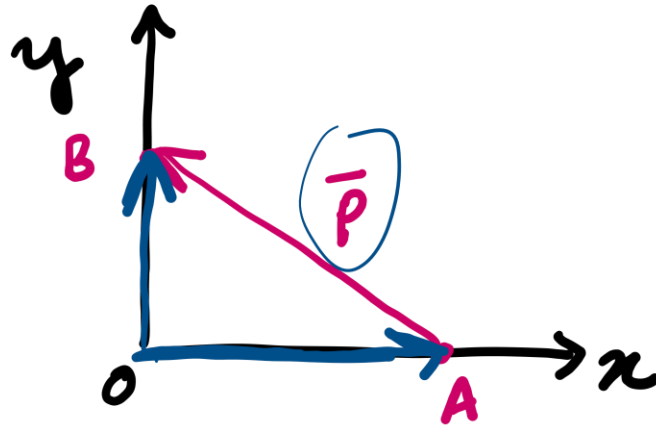
WRITE THE EXPRESSION FOR THE
RELATIONSHIP BETWEEN THE
THREE VECTORS



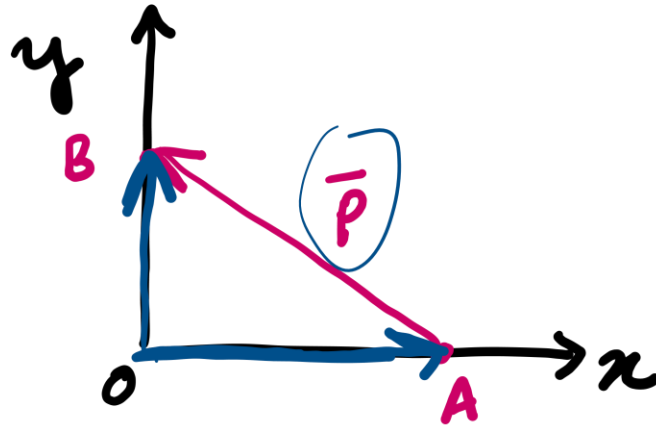
$$\vec{B} + \vec{C} - \vec{A} = 0$$

$$\vec{A} = \vec{B} + \vec{C}$$

PROBLEM ② : WRITE AN EXPRESSION
FOR $\vec{P}$ IN TERMS OF $\vec{OA}$ AND
 $\vec{OB}$



PROBLEM ② : WRITE AN EXPRESSION
FOR $\vec{P}$ IN TERMS OF $\vec{OA}$ AND
 $\vec{OB}$



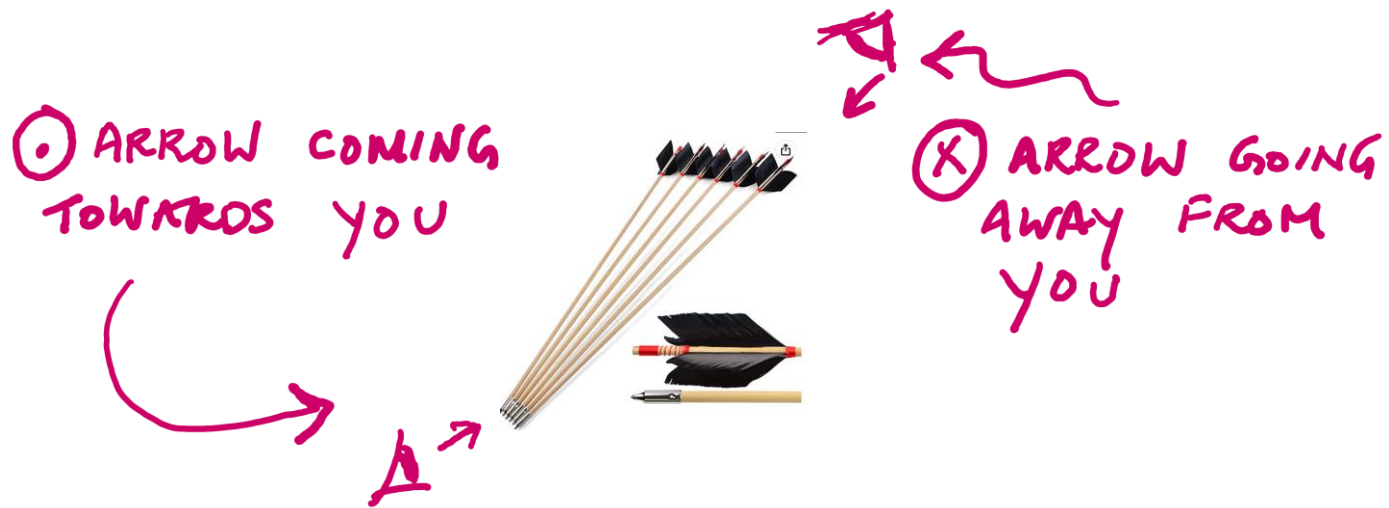
$$\vec{OA} + \vec{P} - \vec{OB} = 0$$

$$\vec{P} = \vec{OB} - \vec{OA}$$

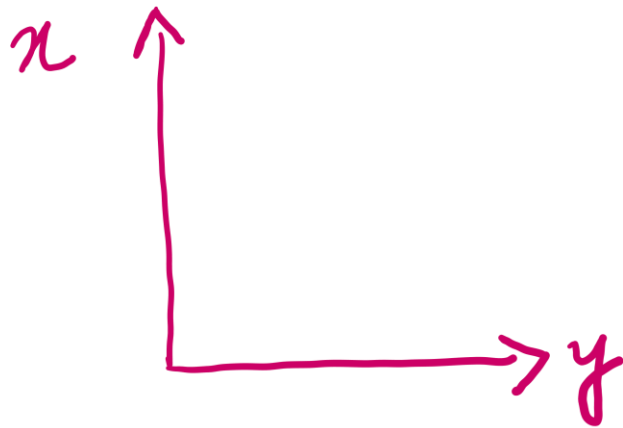
Drawing 3D coordinate system



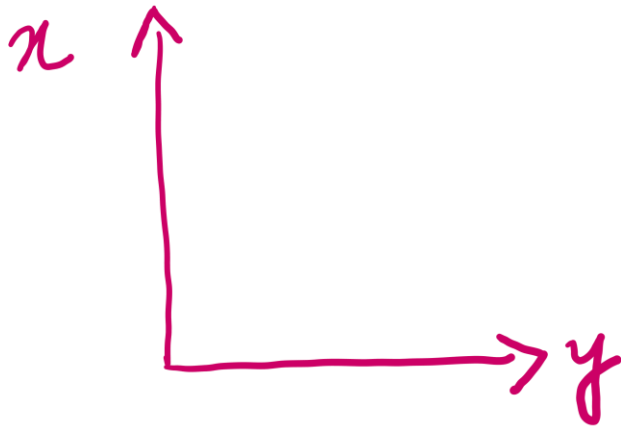
Drawing 3D coordinate system



Q. DRAW THE MISSING AXIS



Q. DRAW THE MISSING AXIS



Example 3-1: Vectors and Angles

In Cartesian coordinates, vector **A** points from the origin to point $P_1 = (2, 3, 3)$, and vector **B** is directed from P_1 to point $P_2 = (1, -2, 2)$. Find

- (a) vector **A**, its magnitude A , and unit vector $\hat{\mathbf{a}}$,
- (b) the angle between **A** and the y -axis,
- (c) vector **B**,
- (d) the angle θ_{AB} between **A** and **B**, and
- (e) the perpendicular distance from the origin to vector **B**.

Solution: (a) Vector **A** is given by the position vector of $P_1 = (2, 3, 3)$ as shown in Fig. 3-7. Thus,

$$\mathbf{A} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3,$$

$$A = |\mathbf{A}| = \sqrt{2^2 + 3^2 + 3^2} = \sqrt{22},$$

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{A} = (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3)/\sqrt{22}.$$

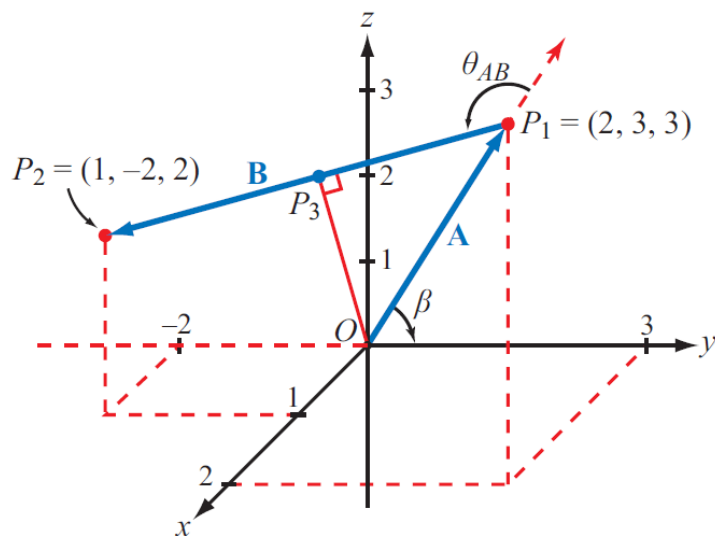


Figure 3-7: Geometry of Example 3-1.

- (b) The angle β between **A** and the y -axis is obtained from

$$\mathbf{A} \cdot \hat{\mathbf{y}} = |\mathbf{A}||\hat{\mathbf{y}}| \cos \beta = A \cos \beta,$$

or

$$\beta = \cos^{-1} \left(\frac{\mathbf{A} \cdot \hat{\mathbf{y}}}{A} \right) = \cos^{-1} \left(\frac{3}{\sqrt{22}} \right) = 50.2^\circ.$$

- (c)

$$\mathbf{B} = \hat{\mathbf{x}}(1 - 2) + \hat{\mathbf{y}}(-2 - 3) + \hat{\mathbf{z}}(2 - 3) = -\hat{\mathbf{x}} - \hat{\mathbf{y}}5 - \hat{\mathbf{z}}.$$

- (d)

$$\begin{aligned} \theta_{AB} &= \cos^{-1} \left[\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|} \right] = \cos^{-1} \left[\frac{(-2 - 15 - 3)}{\sqrt{22} \sqrt{27}} \right] \\ &= 145.1^\circ. \end{aligned}$$

- (e) The perpendicular distance between the origin and vector **B** is the distance $|\overrightarrow{OP_3}|$ shown in Fig. 3-7. From right triangle OP_1P_3 ,

$$\begin{aligned} |\overrightarrow{OP_3}| &= |\mathbf{A}| \sin(180^\circ - \theta_{AB}) \\ &= \sqrt{22} \sin(180^\circ - 145.1^\circ) = 2.68. \end{aligned}$$

Example 3-3: Distance Vector in Cylindrical Coordinates

Find an expression for the unit vector of vector $\mathbf{A}$ shown in Fig. 3-11 in cylindrical coordinates.

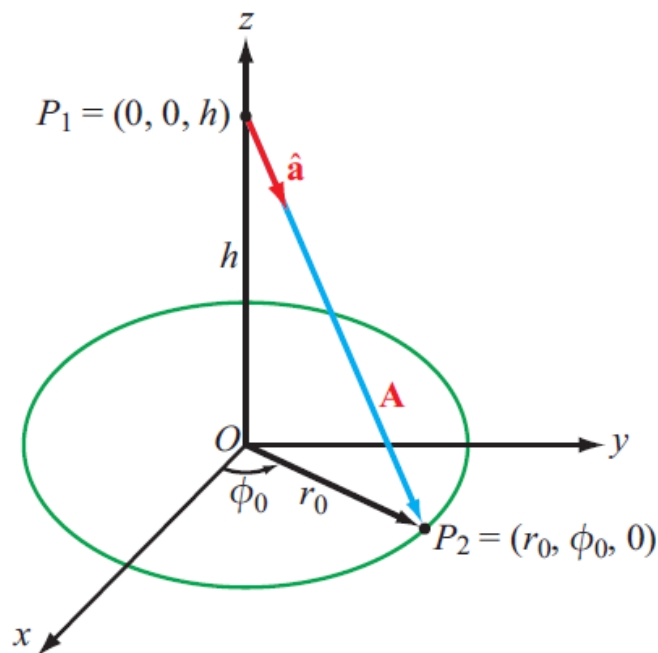


Figure 3-11: Geometry of Example 3-3.

Solution: In triangle OP_1P_2 ,

$$\overrightarrow{OP_2} = \overrightarrow{OP_1} + \mathbf{A}.$$

Hence,

$$\begin{aligned} \mathbf{A} &= \overrightarrow{OP_2} - \overrightarrow{OP_1} \\ &= \hat{\mathbf{r}}r_0 - \hat{\mathbf{z}}h, \end{aligned}$$

and

$$\begin{aligned} \hat{\mathbf{a}} &= \frac{\mathbf{A}}{|\mathbf{A}|} \\ &= \frac{\hat{\mathbf{r}}r_0 - \hat{\mathbf{z}}h}{\sqrt{r_0^2 + h^2}}. \end{aligned}$$

We note that the expression for $\mathbf{A}$ is independent of ϕ_0 . That is, all vectors from point P_1 to any point on the circle defined by $r = r_0$ in the x - y plane are equal in the cylindrical coordinate system. The ambiguity can be eliminated by specifying that $\mathbf{A}$ passes through a point whose $\phi = \phi_0$.

Example 3-5: Surface Area in Spherical Coordinates

The spherical strip shown in Fig. 3-15 is a section of a sphere of radius 3 cm. Find the area of the strip.

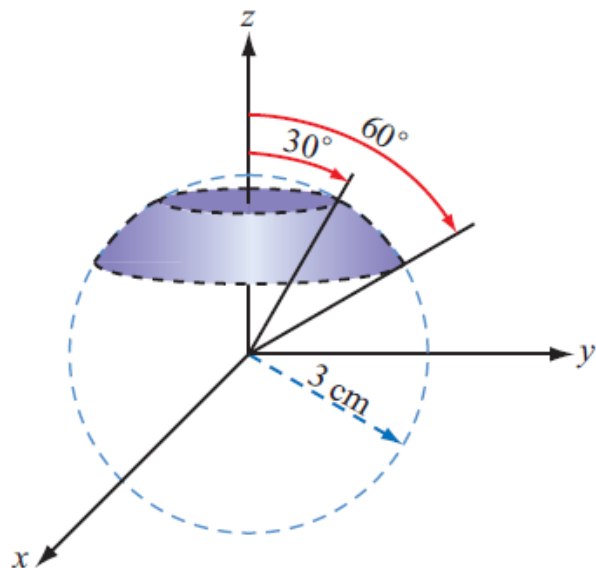


Figure 3-15: Spherical strip of Example 3-5.

Solution: Use of Eq. (3.50b) for the area of an elemental spherical area with constant radius R gives

$$\begin{aligned}
 S &= R^2 \int_{\theta=30^\circ}^{60^\circ} \sin \theta \, d\theta \int_{\phi=0}^{2\pi} d\phi \\
 &= 9(-\cos \theta) \Big|_{30^\circ}^{60^\circ} \phi \Big|_0^{2\pi} \quad (\text{cm}^2) \\
 &= 18\pi(\cos 30^\circ - \cos 60^\circ) = 20.7 \text{ cm}^2.
 \end{aligned}$$

Example 3-6: Charge in a Sphere

A sphere of radius 2 cm contains a volume charge density ρ_v given by

$$\rho_v = 4 \cos^2 \theta \quad (\text{C/m}^3).$$

Find the total charge Q contained in the sphere.

Solution:

$$\begin{aligned}
 Q &= \int_V \rho_v \, dV \\
 &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{R=0}^{2 \times 10^{-2}} (4 \cos^2 \theta) R^2 \sin \theta \, dR \, d\theta \, d\phi \\
 &= 4 \int_0^{2\pi} \int_0^{\pi} \left(\frac{R^3}{3} \right) \Big|_0^{2 \times 10^{-2}} \sin \theta \cos^2 \theta \, d\theta \, d\phi \\
 &= \frac{32}{3} \times 10^{-6} \int_0^{2\pi} \left(-\frac{\cos^3 \theta}{3} \right) \Big|_0^{\pi} d\phi \\
 &= \frac{64}{9} \times 10^{-6} \int_0^{2\pi} d\phi \\
 &= \frac{128\pi}{9} \times 10^{-6} = 44.68 \quad (\mu\text{C}).
 \end{aligned}$$

Example 3-7: Cartesian to Cylindrical Transformations

Given point $P_1 = (3, -4, 3)$ and vector $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}4$, defined in Cartesian coordinates, express P_1 and $\mathbf{A}$ in cylindrical coordinates and evaluate $\mathbf{A}$ at P_1 .

Solution: For point P_1 , $x = 3$, $y = -4$, and $z = 3$. Using Eq. (3.51), we have

$$r = \sqrt{x^2 + y^2} = 5, \quad \phi = \tan^{-1} \frac{y}{x} = -53.1^\circ = 306.9^\circ,$$

and z remains unchanged. Hence, $P_1 = (5, 306.9^\circ, 3)$ in cylindrical coordinates.

The cylindrical components of vector $\mathbf{A} = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$ can be determined by applying Eqs. (3.58a) and (3.58b):

$$A_r = A_x \cos \phi + A_y \sin \phi = 2 \cos \phi - 3 \sin \phi,$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi = -2 \sin \phi - 3 \cos \phi,$$

$$A_z = 4.$$

Hence,

$$\mathbf{A} = \hat{\mathbf{r}}(2 \cos \phi - 3 \sin \phi) - \hat{\boldsymbol{\phi}}(2 \sin \phi + 3 \cos \phi) + \hat{\mathbf{z}}4.$$

At point P , $\phi = 306.9^\circ$, which gives

$$\mathbf{A} = \hat{\mathbf{r}}3.60 - \hat{\boldsymbol{\phi}}0.20 + \hat{\mathbf{z}}4.$$

Example 3-8: Cartesian to Spherical Transformation

Express vector $\mathbf{A} = \hat{\mathbf{x}}(x + y) + \hat{\mathbf{y}}(y - x) + \hat{\mathbf{z}}z$ in spherical coordinates.

Solution: Using the transformation relation for A_R given in Table 3-2, we have

$$\begin{aligned} A_R &= A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta \\ &= (x + y) \sin \theta \cos \phi + (y - x) \sin \theta \sin \phi + z \cos \theta. \end{aligned}$$

Using the expressions for x , y , and z given by Eq. (3.61c), we have

$$\begin{aligned} A_R &= (R \sin \theta \cos \phi + R \sin \theta \sin \phi) \sin \theta \cos \phi \\ &\quad + (R \sin \theta \sin \phi - R \sin \theta \cos \phi) \sin \theta \sin \phi + R \cos^2 \theta \\ &= R \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + R \cos^2 \theta \\ &= R \sin^2 \theta + R \cos^2 \theta = R. \end{aligned}$$

Similarly,

$$\begin{aligned} A_\theta &= (x + y) \cos \theta \cos \phi + (y - x) \cos \theta \sin \phi - z \sin \theta, \\ A_\phi &= -(x + y) \sin \phi + (y - x) \cos \phi, \end{aligned}$$

Using the relations:

$$\begin{aligned} x &= R \sin \theta \cos \phi, \\ y &= R \sin \theta \sin \phi, \\ z &= R \cos \theta. \end{aligned}$$

leads to:

$$\begin{aligned} A_\theta &= 0, \\ A_\phi &= -R \sin \theta. \end{aligned}$$

Hence,

$$\mathbf{A} = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi = \hat{\mathbf{R}}R - \hat{\boldsymbol{\phi}}R \sin \theta.$$